

The Hong Kong Polytechnic University

Department of Computing

COMP4913 Capstone Project

Report (Interim)

From Classic Strategies to Time-Series Transformers: A Quantitative Trading System (Interim Milestone)

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Declaration of Originality

The Hong Kong Polytechnic University
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COMP4913 Capstone Project

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Abstract

This interim report documents my first end-to-end milestone of a quantitative algorithmic trading system prototype using 2800.HK daily OHLCV data. I begin with classic, interpretable strategies to validate the research loop and set a transparent baseline. I then implement a consistent ML/DL pipeline: feature engineering, volatility-adaptive 3-class labeling, fixed-window walk-forward cross-validation, and a cost-aware backtest that converts out-of-sample class probabilities into trades. Under the same protocol, I compare an LSTM baseline, a Transformer encoder baseline, and an Informer-style time-series Transformer, and I show that realized P&L is often dominated by the probability-to-trade translation layer (thresholds, holding constraints, and costs). I further report an Informer-OPT result obtained via feature expansion, longer context, and a trading-rule parameter sweep. Finally, I integrate a news-based stage (GDELT + FinBERT) as an external signal pipeline; this stage is currently a systems milestone rather than a performance conclusion due to limited news coverage and short out-of-sample overlap.

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Chapter 1

Introduction

1.1 Motivation

Financial markets are noisy and shaped by many interacting factors. While the Efficient Market Hypothesis suggests persistent alpha is difficult [Fama 1970], practical research can still explore short-horizon signals, provided the evaluation is time-respecting and reproducible.

1.2 Midterm objective and scope

My midterm objective is to build a complete trading research pipeline that is (i) time-respecting (no look-ahead leakage), (ii) reproducible, and (iii) able to generate standardized artifacts (metrics, plots, logs) for iterative experiments. This report presents a progressive path: classic strategies → LSTM → Transformer → Informer-style → Informer-OPT → news integration (FinBERT).

Chapter 2

Problem Definition and Data

2.1 Instrument and data source

Instrument: 2800.HK (Tracker Fund of Hong Kong), chosen for liquidity and relevance as a Hang Seng Index proxy.

Data source: Daily OHLCV from `yfinance` with auto-adjusted prices.

2.2 Time span and evaluation window

Most early experiments use three years of daily bars (period=3y, interval=1d), yielding 733 observations. When optimizing Informer with longer context windows, I expanded the history request to period=10y to preserve enough out-of-sample folds. For ML/DL models, headline metrics are reported on the final 252 out-of-sample trading days produced by walk-forward CV for that run.

2.3 Prediction target

I predict next-day close-to-close log return:

$$r_{t+1} = \log \left(\frac{C_{t+1}}{C_t} \right).$$

Chapter 3

Methodology: A Reproducible, Time-Respecting Pipeline

3.1 Feature engineering

I begin with 16 interpretable technical features: SMA(5/10/20/50), rolling variance of log returns (5/10/20), RSI(14), MACD (line/signal/histogram), and Bollinger band features (mid/upper/lower/bandwidth/%). For Informer-OPT, I expand the feature set with multi-horizon returns/volatilities, ATR-based volatility, and volume-derived signals.

3.2 Volatility-adaptive 3-class labeling

I use a dynamic threshold tied to recent volatility to reduce regime sensitivity. Let $\sigma_t^{(20)}$ be the rolling standard deviation of 1-day log returns over a 20-day window, shifted by one day to avoid leakage. The threshold is:

$$\tau_t = \max(\text{min_vol}, k \cdot \sigma_t^{(20)}), \quad k = 0.8, \text{ min_vol} = 10^{-4}.$$

Labels: Up if $r_{t+1} \geq \tau_t$, Down if $r_{t+1} \leq -\tau_t$, else Neutral.

3.3 Walk-forward cross-validation

I use fixed-window walk-forward splits with train/validation/test windows and step size chosen to avoid lookahead leakage. Standardization is fit on each training split only and applied to validation/test.

3.4 From probabilities to trades (cost-aware backtest)

For each out-of-sample day, the model outputs probabilities ($P(\text{Down})$, $P(\text{Neutral})$, $P(\text{Up})$). I apply a confidence threshold: if $\max P < \text{thr}$, I force Neutral (no trade). Otherwise, I map to a

discrete position (long/short) and enforce a minimum holding period to reduce flip-flopping. Transaction cost is charged on absolute position changes at 2 bps. This “translation layer” is evaluated explicitly because it often dominates realized P&L [Bailey et al. 2014, López de Prado 2018].

Chapter 4

Classic Strategy Baselines (Pre-ML)

4.1 Why classic baselines first

Before ML/DL, I implemented classic, interpretable strategies to validate the data/backtest loop and build intuition for 2800.HK [Chan 2013]. These baselines are not optimized; they provide a transparent benchmark.

4.2 Representative results (3-year sample)

Table 4.1 summarizes representative classic strategies (3-year sample, costs included).

Strategy	Total ret.	Sharpe	Max DD	Coverage
RSI	0.1325	1.2977	-0.0029	0.0109
Bollinger (MR)	0.0735	0.4944	-0.0998	0.0464
Dual MA (10/50)	0.1075	0.2864	-0.2010	0.4911
Donchian (20/10)	-0.0275	0.0047	-0.2164	0.3438
Buy&Hold (same 3y)	0.4436	–	–	1.0000

Table 4.1: Classic strategy baselines on 2800.HK (full 3-year sample; 2 bps transaction cost). Some strategies show strong risk-adjusted metrics but trail Buy&Hold in total return due to low exposure and trading frictions.

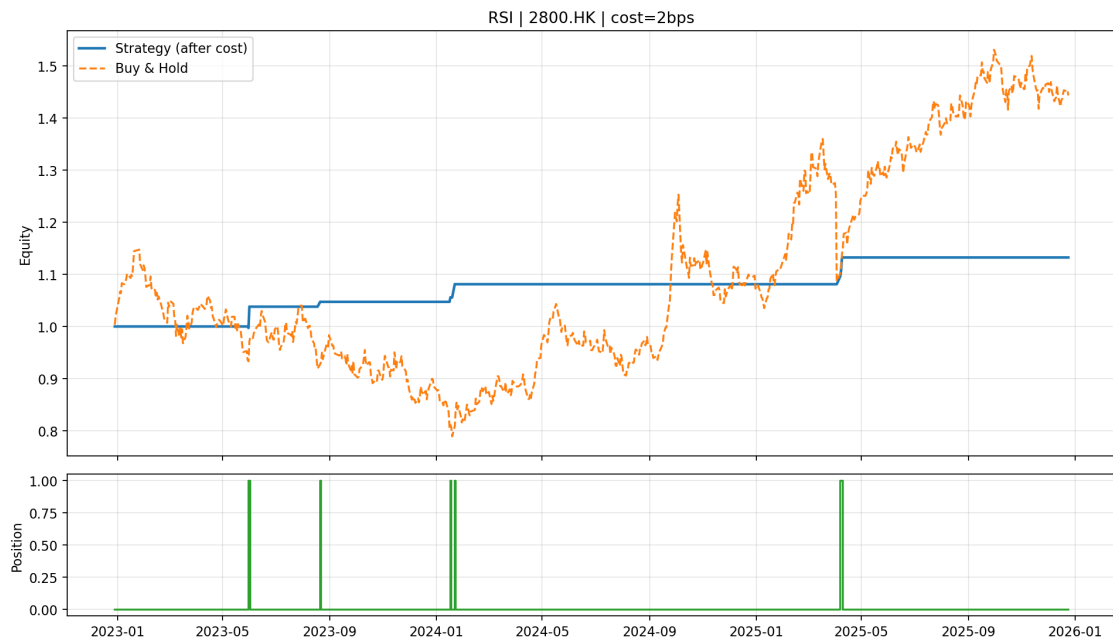


Figure 4.1: Classic baseline example: RSI mean-reversion (equity curve and position).

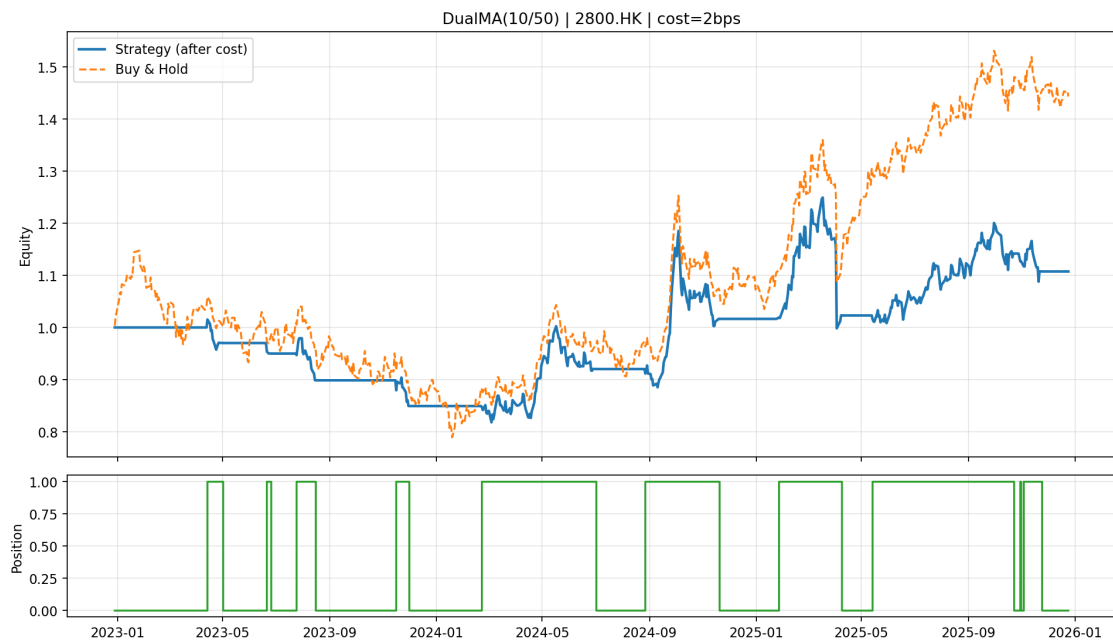


Figure 4.2: Classic baseline example: Dual moving-average trend (equity curve and position).

Chapter 5

ML/DL Experiments Under a Consistent Protocol

5.1 LSTM baseline (first reproducible ML run)

I train a PyTorch LSTM multi-class classifier with lookback=30 and 16 features. The run produces standardized artifacts: configuration snapshot, walk-forward CV metrics, per-day probabilities, and a cost-aware backtest on the last 252 out-of-sample trading days.

Metric	LSTM Strategy	Buy&Hold
Days	252	252
Total return	-0.0623	0.2230
Annualized return	-0.0623	0.2230
Annualized volatility	0.1033	–
Sharpe ratio	-0.5711	–
Max drawdown	-0.1344	–
Position coverage	0.2540	1.0000

Table 5.1: Backtest summary for the LSTM baseline (last ~1 year out-of-sample; costs included).

5.2 Transformer encoder baseline (GPU run)

I implement a Transformer encoder classifier and evaluate it under the same protocol [Vaswani et al. 2017]. This increases model capacity and enables stronger sequence modeling without recurrence.

5.3 Informer-style time-series Transformer

I implement an Informer-inspired encoder classifier with ProbSparse attention and optional distilling to explore efficient long-sequence mechanisms [Zhou et al. 2021]. Early trials highlight that the trading-rule parameters (confidence threshold and holding period) can suppress trading or con-

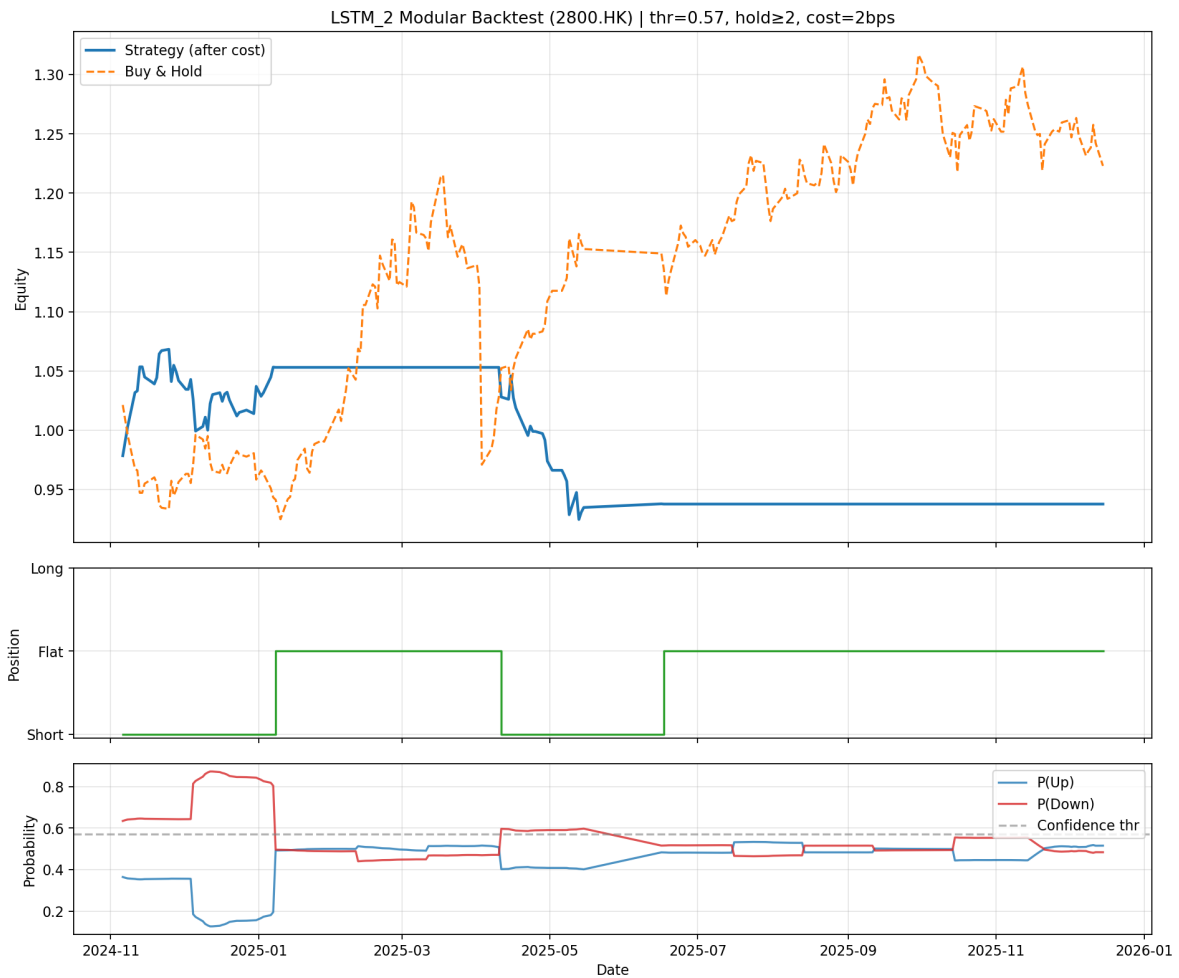


Figure 5.1: LSTM backtest visualization (equity, position, probabilities).

centrate exposure unintentionally; therefore, I treat the trading layer as part of the evaluated system [Bailey et al. 2014, López de Prado 2018].

5.4 Informer-OPT: feature expansion + longer context + trading-rule sweep

I then perform a structured optimization: expand features, increase temporal context, switch to a full 3-class loss for better calibration, and sweep the trading-rule parameters without retraining the model. The best configuration from the sweep nearly matches Buy&Hold over the same window with lower drawdown and modest turnover.

Metric	Transformer Strategy	Buy&Hold
Days	252	252
Total return	0.0094	0.2230
Annualized return	0.0094	0.2230
Annualized volatility	0.2173	–
Sharpe ratio	0.1482	–
Max drawdown	-0.1732	–
Position coverage	0.6230	1.0000

Table 5.2: Backtest summary for the Transformer baseline (last ~1 year out-of-sample; costs included).

Metric	Informer-style	Buy&Hold
Days	252	252
Total return	-0.0102	0.2230
Sharpe ratio	-0.0119	–
Max drawdown	-0.1528	–
Position coverage	0.4206	1.0000
Turnover (avg $ \Delta pos $)	0.0595	–
Trades (pos changes)	14	–

Table 5.3: Backtest summary for the Informer-style baseline (last ~1 year out-of-sample; costs included).

Metric	Informer-OPT (best)	Buy&Hold
Days	252	252
Total return	0.3675	0.3692
Excess return	-0.0017	–
Annualized volatility	0.1677	–
Sharpe ratio	1.9501	–
Max drawdown	-0.0751	–
Position coverage	0.6905	1.0000
Trades (pos changes)	7	–
Trading rule	thr=0.36, hold $\geq$ 5, long-only	–

Table 5.4: Informer-OPT best result from trading-rule sweep (trading_param_sweep_20251229_234600.csv).

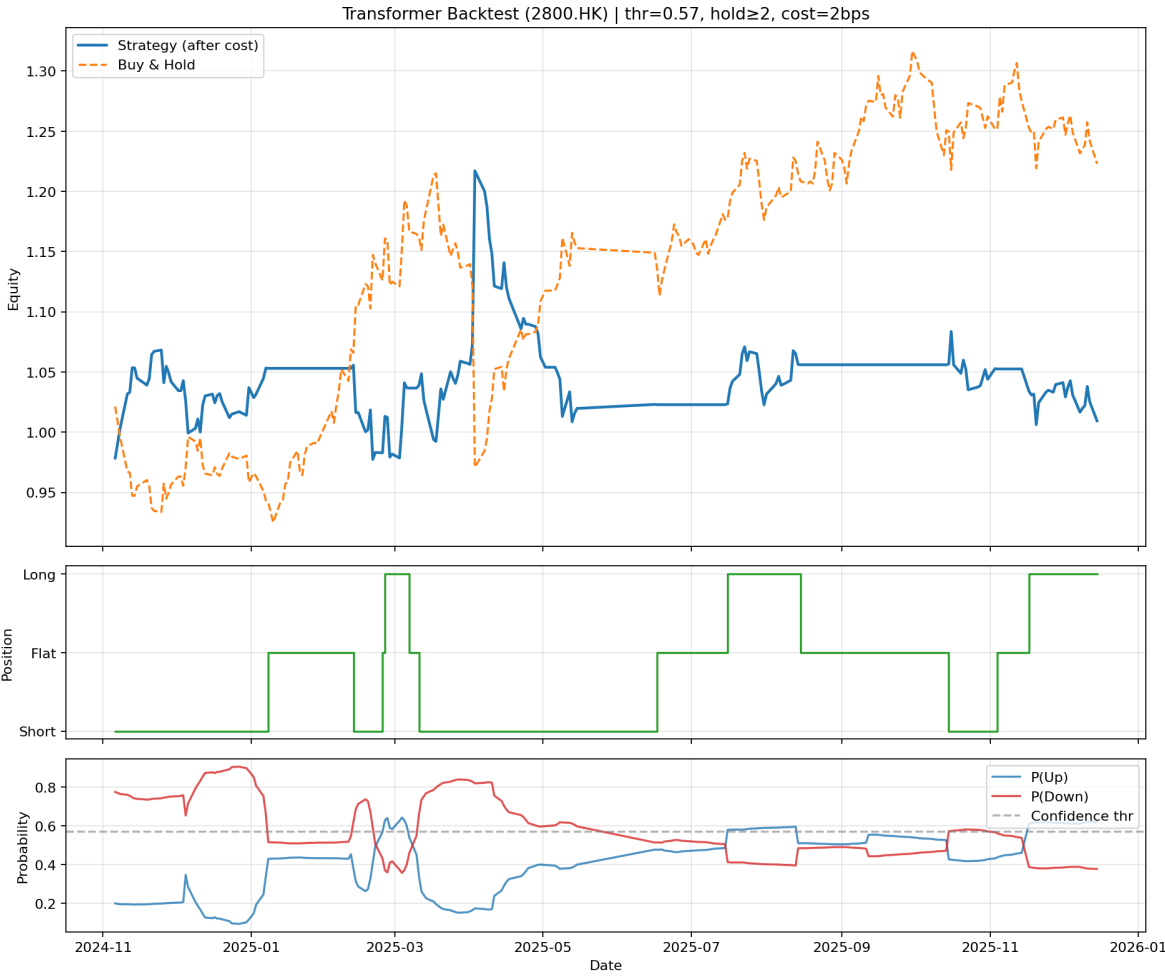


Figure 5.2: Transformer backtest visualization (equity, position, probabilities).

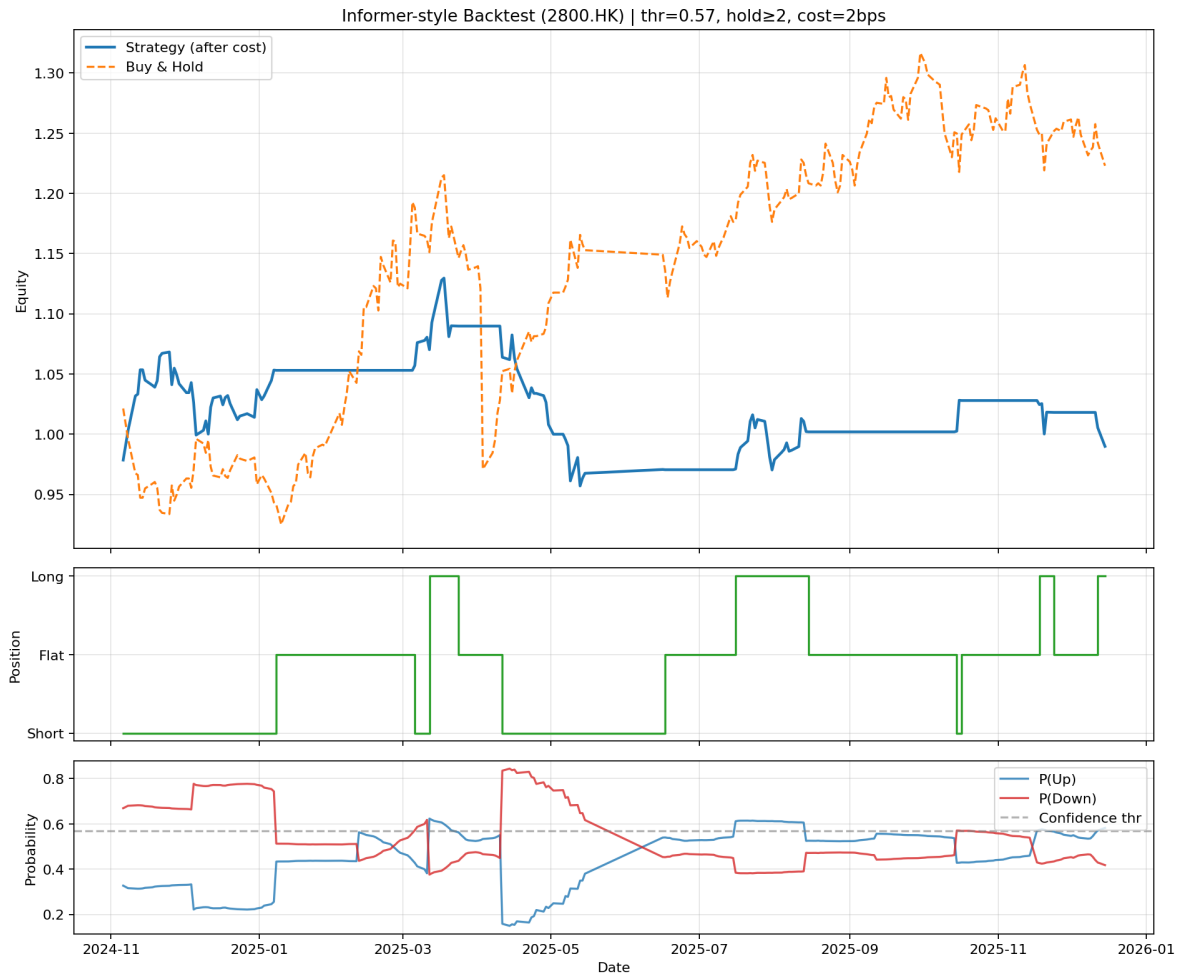


Figure 5.3: Informer-style backtest visualization (equity, position, probabilities).

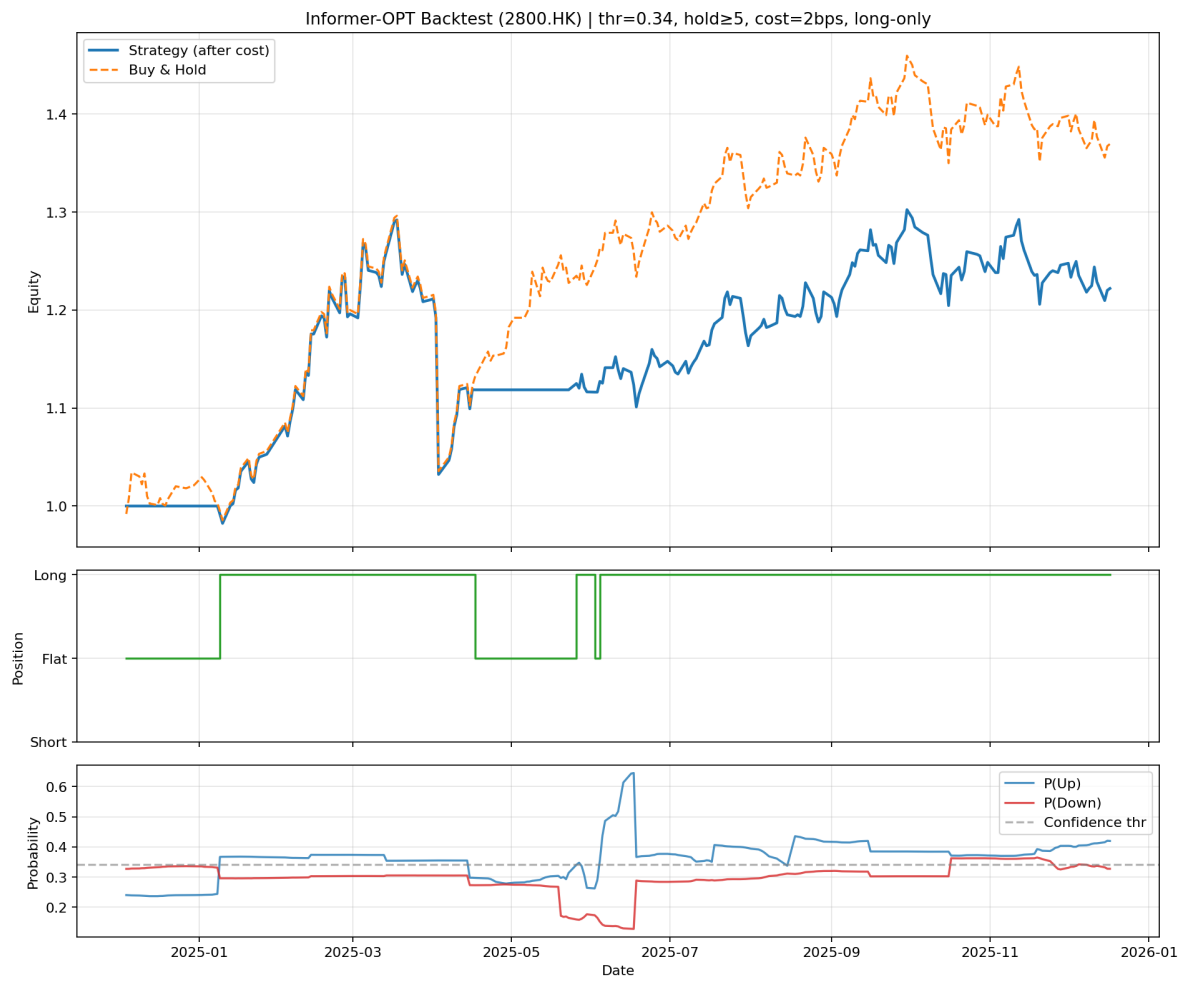


Figure 5.4: Informer-OPT backtest visualization (equity, position, probabilities).

Chapter 6

News as an External Signal (FinBERT Integration)

6.1 Motivation: why news, and why it relates to 2800.HK

Price-only signals may miss external drivers. Because 2800.HK tracks a broad HK equity index, macro and market news can influence next-day sentiment and flows. In this stage, my goal is to build a scalable news-to-trade workflow that can be evaluated once coverage is sufficient.

6.2 Data acquisition (GDELT) and preprocessing

I use the GDELT 2.1 Document API to fetch news, map it to HK tickers via alias queries, and aggregate text by (date,ticker). The processed dataset is stored as `data/news/news.csv` with schema `date,ticker,text`.

6.3 Pretrained model and fine-tuning objective

I start from FinBERT (`ProsusAI/finbert`) [Araci 2019] and attempt to fine-tune it to predict next-day direction labels (Down/Neutral/Up), treating the output as a news-sentiment/news-signal classifier. I then convert probabilities to positions and backtest an equal-weight HK portfolio.

6.4 Interim status

The pipeline runs end-to-end and produces alignment diagnostics, but the out-of-sample window is currently very short due to limited overlap between news and price dates. Therefore, this stage is reported as a systems milestone rather than a performance conclusion.

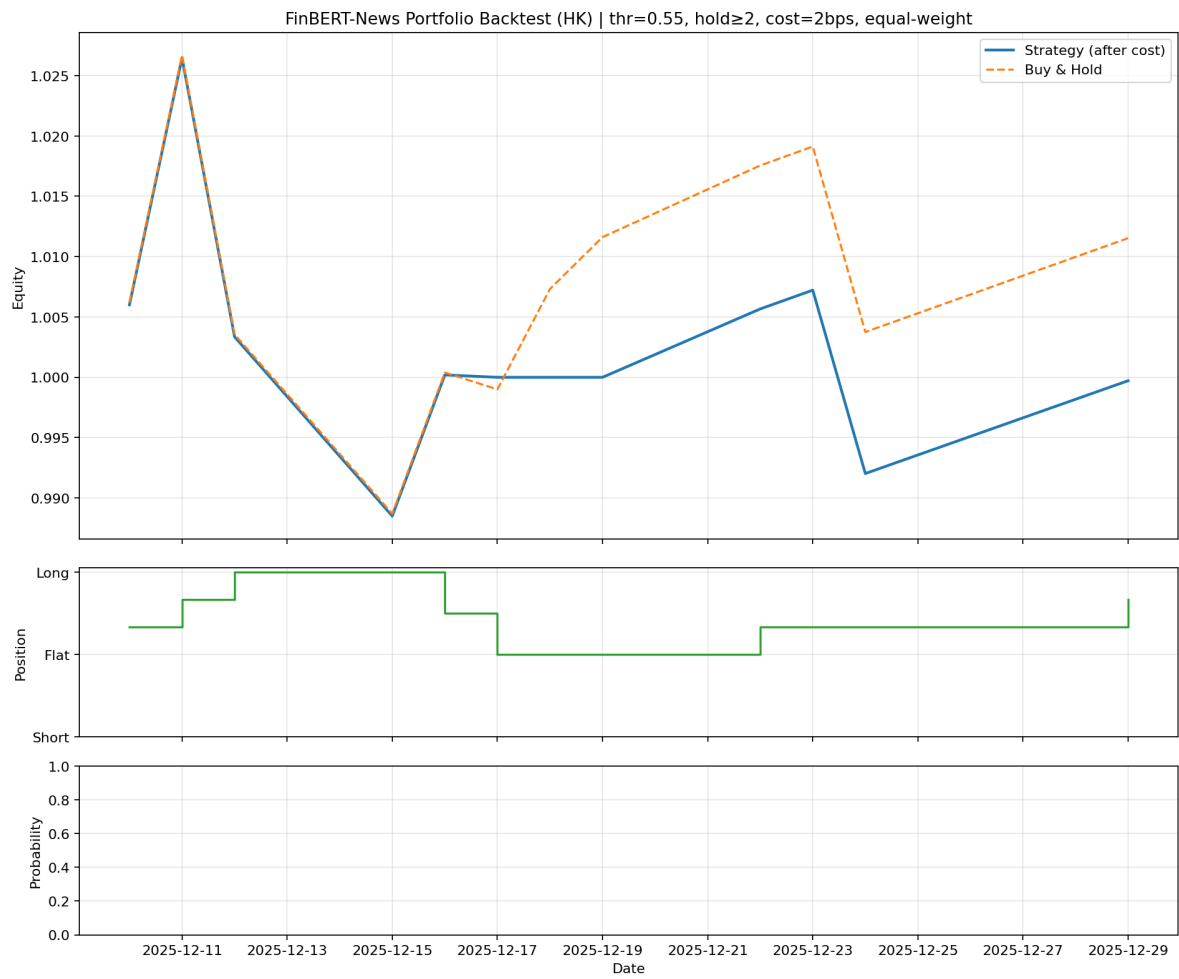


Figure 6.1: FinBERT-news portfolio backtest visualization (short window due to limited news coverage).

Chapter 7

Progress Summary and Lessons Learned

7.1 What I have achieved so far

By midterm, I have achieved the following deliverables:

- A reproducible pipeline that produces standardized artifacts in `outputs/` (configs, CV metrics, per-day predictions, backtest statistics, and plots).
- Classic strategy baselines that validate the backtest mechanics and provide transparent benchmarks.
- ML/DL baselines (LSTM, Transformer, Informer-style) evaluated under a consistent time-respecting protocol.
- An explicit optimization step (Informer-OPT) that demonstrates the importance and risk of the probability-to-trade translation layer.
- A first integrated news stage (GDELT + FinBERT) with data alignment diagnostics, setting up the next research phase.

7.2 Key lessons

Time-respecting evaluation is non-negotiable [3]. In addition, backtests can be overfit at the trading-rule layer even without retraining the model [Bailey et al. 2014, López de Prado 2018]. Finally, data coverage and alignment constrain what can be concluded from news-based modeling.

Chapter 8

Work Plan (Next Stage)

To progress from a systems milestone to a defensible research result, my next steps are:

- Expand news coverage (longer time span, broader HK entities/tickers) [Leetaru and Schrodtt 2013].
- Stabilize fine-tuning and evaluate with longer out-of-sample windows.
- Extend multi-asset evaluation and portfolio-level decisions under consistent costs.
- Continue exploring time-series Transformers with careful separation between model training and downstream decision calibration.

Chapter 9

Conclusion

This interim report documents a complete, time-respecting trading research pipeline and a sequence of progressively stronger baselines. While most price-based ML/DL runs do not beat Buy&Hold in total return, the experiments establish a reproducible evaluation framework and reveal that the probability-to-trade translation layer is a dominant driver of realized performance. The news stage (GDELT + FinBERT) is integrated and debuggable, and the next phase is to expand coverage so that text-based signals can be evaluated fairly.

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Appendix A

Handbook Compliance Checklist (Interim Report)

- Organization: cover page, abstract, TOC, lists, main body, references, appendices.
- Cover content: title, student ID, programme stream, supervisor, date; personal details appear only on cover.
- Formatting: 12pt Times-like font, at least 1.5 spacing; margins left 25mm, top/bottom/right 20mm; A4.
- Pagination: sequential, top-right; title page counted but not numbered.
- Length: normally not exceeding 50 pages excluding preamble and appendices.
- Submission: PDF; maximum file size 50MB.